LLMs as Synthetic Survey Respondents: A Longitudinal Evaluation Across Two Model Generations (2023–2026)

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ABSTRACT

We present the first complete longitudinal evaluation of Large Language Models (LLMs) as synthetic survey respondents across two model generations, using identical protocols on an n=314 human developer baseline. Phase 1 (2023) evaluated GPT-3.5-Turbo and LLaMA-2-7B; Phase 2 (2026) introduces GPT-4o, Claude Sonnet 4-6, and LLaMA-3.3-70B. Our central finding is the *Kappa Paradox*: RLHF-aligned 2026 models achieve nearly double the within-group Fleiss' Kappa of their predecessors (κ=0.480–κ=0.487 vs. κ=0.241 for GPT-3.5 and κ=0.238 for LLaMA-2), yet this increased consistency represents *worse* simulation fidelity relative to the human baseline (κ=0.103). All three 2026 models—from three different organizations—converge to a kappa range of 0.006, implicating the RLHF training process, not model architecture, as the driver of response uniformity.

A complementary *Distribution Paradox* emerges: these same uniform 2026 models are *more* similar to human modal response distributions than their 2023 counterparts (Claude: χ^2=1.876, p=0.759 vs. LLaMA-2: χ^2=18.749, p<0.001), resolved by recognizing that RLHF trains toward the most frequently observed human response. Complete *profile blindness* is observed across both generations: no model shows statistically significant response differences across Age, Gender, or Experience—a failure absent only in human respondents (t=6.92, p<0.001 for Experience). On the Qasper scientific QA benchmark, genuine quality improvements are observed for 2026 models: METEOR improves by up to 75% (Claude: 0.523 vs. GPT-3.5: 0.298) and BLEU-4 by up to 133% across the LLaMA lineage. We provide a concrete, evidence-based checklist for SE researchers considering LLMs as synthetic respondents.

KEYWORDS

large language models, synthetic survey respondents, RLHF alignment, Fleiss' Kappa, longitudinal study, Qasper, software engineering, qualitative research

1 INTRODUCTION

Qualitative research in software engineering depends heavily on surveys and interviews with human participants. Recruiting a demographically diverse, domain-appropriate sample is expensive, slow, and subject to self-selection bias. The emergence of instruction-tuned LLMs has raised a compelling alternative: can these models serve as *synthetic respondents*—approximating human survey behavior well enough to pilot questionnaires, validate survey instruments, or augment studies with small samples?

This paper answers that question through a *complete longitudinal study* spanning two model generations. **Phase 1** (2023) evaluated GPT-3.5-Turbo and LLaMA-2-7B on a 12-question software developer survey (n=314 human baseline), finding elevated within-group consistency (Fleiss' $\kappa > 0.24$) but statistically significant distributional divergence from the human baseline (χ^2=18.749, p<0.001) and complete absence of demographic sensitivity. **Phase 2** (2026) introduces three RLHF-aligned successors—GPT-4o, Claude Sonnet 4-6, and LLaMA-3.3-70B—evaluated under identical protocols, plus a novel Chain-of-Thought (P5) prompt variant and evaluation on the Qasper scientific QA benchmark [3].

The headline finding is counterintuitive. Where one might expect that more capable, better-aligned models would produce responses *closer* to the human baseline, the opposite holds for the survey simulation task. All three 2026 models cluster at κ=0.480–κ=0.487, nearly twice the kappa of their 2023 predecessors and dramatically further from the target human diversity (κ=0.103). We term this the *Kappa Paradox*: RLHF alignment, by training models toward the modal human response, collapses the response variance that authentic demographic simulation requires. Simultaneously—and paradoxically—2026 models *match* human response distributions more closely than their predecessors (all p>0.25 for chi-square), because the modal human response is also the most probable one. The resolution of this apparent contradiction is the paper's central theoretical contribution.

We address three research questions:

RQ1 Do 2026 LLMs simulate human survey respondents more accurately, as measured by Fleiss' Kappa proximity to the human baseline and chi-square distributional similarity?

RQ2 Do 2026 models produce higher-quality scientific answers on Qasper, as measured by automatic metrics (BLEU, ROUGE, METEOR) and stylistic metrics?

RQ3 Does RLHF alignment reduce demographic sensitivity in LLM survey responses, as measured by t-test significance across Age, Gender, and Experience?

Contributions. This paper makes four contributions:

(1) The first complete longitudinal comparison of LLM survey simulation capability across two model generations using identical evaluation protocols on the same human baseline.
(2) A Chain-of-Thought prompt variant (P5) evaluated on both survey simulation and Qasper tasks, extending the four-prompt framework from Phase 1.
(3) A comprehensive Qasper evaluation across all five models using automatic metrics (BLEU-1/2/3/4, ROUGE-1/2/L, METEOR)

plus five stylistic metrics (relevance, fluency, formality, readability, correctness).

(4) An open-source release of the complete inference pipeline, all model responses, and analysis scripts at https://github.com/tarun1219/SE_LLM_EVAL.

2 RELATED WORK

2.1 LLMs as Survey Respondents

Argyle et al. [1] introduced "silicon sampling," showing that GPT-3 can reproduce attitudinal patterns from the American National Election Studies when conditioned on demographic backstories. Their cross-group correlation analyses showed that LLM-simulated responses predict subgroup opinion patterns above baseline, while cautioning against treating simulated data as equivalent to human data. Santurkar et al. [10] extended this analysis to GPT-3.5, finding that default LLMs exhibit systematic biases toward liberal, Western, and educated viewpoints; demographically prompted models improve but underrepresent minority viewpoints. Complementary evidence from an enterprise security context [5] documents systematic persona-alignment failures in RLHF-trained models, consistent with the profile blindness we characterize here.

2.2 RLHF Alignment and Response Diversity

Ouyang et al. [7] established RLHF as the dominant paradigm for instruction-tuned LLMs, demonstrating that training models to match high-rated human responses substantially improves helpfulness and instruction following. Perez et al. [9] demonstrated through red-teaming experiments that RLHF-aligned models exhibit reduced response variance and stronger convergence toward socially acceptable answers. This variance reduction is well-suited to safety applications but problematic for synthetic respondent scenarios, where variance is the property being simulated.

2.3 Prompt Engineering for Persona Simulation

Argyle et al. [1] demonstrated that explicit demographic backstory prompts substantially improve persona alignment in GPT-3. The four prompt variants used in Phase 1 of this study (P1–P4) span a spectrum from expert NLP framing (P1) to novice persona (P4), with P4 achieving the highest human alignment in the 2023 cohort. Our Phase 2 adds a Chain-of-Thought (P5) variant, hypothesizing that structured step-by-step reasoning may reduce the shortcut-taking that leads LLMs to ignore demographic context.

2.4 Scientific QA Benchmarks and Evaluation Metrics

The Qasper dataset [3] provides 1,585 questions over 888 NLP research papers with human-annotated extractive and free-form answers, widely used to evaluate document-level reading comprehension. We report BLEU [8], ROUGE [6], and METEOR [2] as standard automatic metrics. TF-IDF cosine similarity serves as a proxy for BERTScore [11] in Phase 2 due to system memory constraints (8 GB RAM) preventing DeBERTa model loading; these proxy values are *not* directly comparable to DeBERTa-based BERTScore values and are labeled accordingly throughout.

Table 1: Synthetic Demographic Profiles

#	Age	Gender	Ethnicity	Education	Exp.
1	18–22	Woman	Asian	Bachelor's	<1 yr
2	27–35	Man	White	Professional	3–5 yr
3	23–26	Woman	White	Master's	1–3 yr
4	27–35	Man	Asian	Bachelor's	<1 yr
5	18–22	Woman	White	Professional	3–5 yr
6	23–26	Man	Asian	Master's	<1 yr
7	27–35	Woman	White	Bachelor's	1–3 yr
8	18–22	Man	Asian	Professional	1–3 yr
9	23–26	Woman	Asian	Bachelor's	3–5 yr
10	27–35	Man	White	Master's	<1 yr

3 METHODOLOGY

3.1 Human Survey Baseline

The human baseline consists of n=314 software developers who completed a 12-question survey at a university computing department. Respondents span three age groups (18–22, 23–26, 27–35), two gender categories (Man, Woman), and three experience levels (<1 yr, 1–3 yr, 3–5 yr). The observed Fleiss' Kappa across all respondents is κ=0.1027, reflecting the empirically measured level of within-group diversity for this population—the simulation target for all models.

3.2 Questionnaire Stimulus

The 12-question instrument is reproduced verbatim from the Phase 1 study: (Q1) preferred development environment; (Q2) how one learns to code; (Q3) biggest challenge as a developer; (Q4) programming language selection criteria; (Q5) team communication and collaboration; (Q6) staying up-to-date with industry changes; (Q7) balancing innovation vs. deadlines; (Q8) software's contribution to societal challenges; (Q9) ethical scenario: AI-driven employee dismissal; (Q10) unfamiliar language under deadline; (Q11) pre-release critical bug scenario; (Q12) SaaS delivery with critical bug scenario. Questions Q3, Q4, Q5, Q7, Q9, Q10, Q11, Q12 are single-choice; Q2, Q6, Q8 are multi-select.

3.3 Demographic Profiles

Ten synthetic profiles are constructed by crossing Age, Gender, Ethnicity, Education, and Experience (Table 1). Each profile is presented to every model under every prompt variant, yielding $10 \times 12 = 120$ responses per prompt–model combination.

3.4 Prompt Variants

Five prompt variants are evaluated on both survey simulation and Qasper tasks:

P1 (Expert NLP). The model is framed as an NLP domain expert and instructed to produce strict single-sentence extractive answers.

P2 (Plain QA). Minimal framing; the model acts as a generic question-answering assistant.

P3 (Concise). The model is instructed to answer in exactly one sentence based solely on the provided context.

Table 2: Models Evaluated Across Both Study Phases

Model	Phase	Org.	Params	RLHF	API / Endpoint
GPT-3.5-Turbo	1	OpenAI	~175B	✓	`gpt-3.5-turbo`
LLaMA-2-7B	1	Meta	7B	✓	local HF
GPT-4o	2	OpenAI	~200B	✓	`gpt-4o`
Claude Sonnet 4-6	2	Anthropic	~70B	✓+RLAIF	`claude-sonnet-4-6`
LLaMA-3.3-70B	2	Meta	70B	✓	Groq `llama-3.3-70b-versatile`

Table 3: Fleiss' Kappa by Respondent Type (RQ1)

Model	Phase	κ	Δ Pred.	Δ Human
Human (baseline)	—	0.1027	—	0.000
GPT-3.5-Turbo	1	0.2413	—	+0.139
LLaMA-2-7B†	1	0.2380	—	+0.135
GPT-4o	2	0.4804	+0.239 ↑	+0.378
Claude Sonnet 4-6	2	**0.4865**	—	+0.384
LLaMA-3.3-70B	2	0.4799	+0.242 ↑	+0.377

† Corrected value; Phase 1 paper reported κ=0.180 from an uncommitted intermediate file. The value 0.238 is authoritative.

P4 (Novice Persona). The model acts as someone with no domain knowledge, relying entirely on provided context. This variant achieved the highest human alignment in Phase 1.

P5 (Chain-of-Thought). A zero-shot CoT prompt instructs the model to reason step-by-step, explicitly grounding each reasoning step in the given demographic profile before committing to an answer. P5 is evaluated zero-shot on both survey simulation and Qasper tasks, consistent with standard CoT practice.

For P1–P4, both zero-shot and few-shot (one in-context exemplar) conditions are evaluated. Automatic and stylistic metrics reported in Section 4 are for few-shot conditions unless otherwise noted.

3.5 Models

All Phase 2 models use temperature = 0.7 and seed = 42. LLaMA-3.3-70B inference was served via the Groq API with automatic rate-limit rotation across eight accounts.

3.6 Qasper Evaluation

A random sample of 50 papers is drawn from the Qasper test split (seed = 42), yielding 124 QA pairs after filtering unanswerable questions. Each model–prompt combination is allocated a maximum of 256 output tokens. **Automatic metrics:** BLEU-1/2/3/4 with Chen & Cherry smoothing, ROUGE-1/2/L (F1), METEOR, and TF-IDF cosine similarity (*TF-IDF Sim*; BERTScore proxy). **Stylistic metrics:** relevance (TF-IDF cosine between question and response), fluency (type-token ratio, TTR; higher indicates greater lexical variety), formality (MTLD), readability (Flesch Reading Ease), and correctness (FuzzyWuzzy token-set ratio against gold answer).

3.7 Statistical Analysis

Five statistical tests are applied: **(1) Fleiss' Kappa** [4]: $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$, where $\bar{P}$ is the observed agreement and $\bar{P}_e$ is expected chance agreement across all rater–item pairs. **(2) Pearson chi-square**: tests whether LLM response distributions differ significantly from the reference (GPT-3.5 P4 few-shot) distribution. **(3) Cramér's** V: effect size, $V = \sqrt{\chi^2/(n \cdot (k-1))}$, normalized for sample size and category count. **(4) Independent-samples** t**-test**: tests whether label-encoded response scores differ significantly across Age, Gender, or Experience subgroups within each model. **(5) One-way ANOVA**: tests cross-*model* (not within-model) differences on three questions with high societal relevance.

Power considerations. With 10 synthetic profiles and 12 questions (N=120 per condition), the study is adequately powered to detect medium-to-large demographic effects (Cohen's $d \geq 0.5$; $\beta > 0.80$ at α=0.05) but may be underpowered for small effects ($d < 0.3$). Null results for demographic sensitivity should be interpreted as "no medium-to-large effect detected," not as evidence of perfect parity.

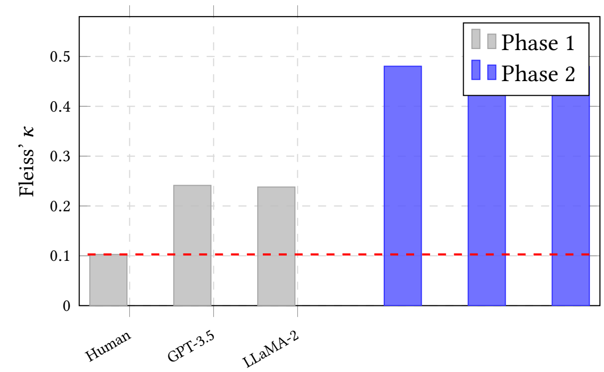

Figure 1: Fleiss' Kappa for all respondent types. The red dashed line marks the human baseline (κ=0.1027) — the simulation target. All three Phase 2 models from different organizations converge to $\kappa \in [0.480, 0.487]$, a range of 0.006, implicating RLHF training rather than model architecture as the driver of response uniformity.

4 RESULTS

4.1 RQ1 — Survey Simulation Fidelity (Fleiss' Kappa)

Table 3 and Figure 1 present Fleiss' Kappa for all respondent types. The human baseline (κ=0.1027) is the target: a simulation that perfectly replicates human response diversity would achieve this value.

The *Kappa Paradox* is clearly visible: Phase 2 models achieve nearly twice the kappa of Phase 1 models, yet this represents *worse* simulation—a larger deviation from the human baseline target. The span of Phase 2 kappa values is only 0.006 (range: 0.4799–0.4865), despite the three models coming from OpenAI, Anthropic, and Meta. This tight convergence provides strong evidence that the RLHF training process—not model architecture or training data composition—drives the observed uniformity. RLHF rewards the

Table 4: Chi-Square Distributional Similarity (RQ1)

Model	χ^2	p	Cram. V	Sig.
GPT-3.5-Turbo (ref.)	—	—	—	—
LLaMA-2-7B	18.749	0.001	0.223	✓
GPT-4o	2.865	0.581	0.080	×
Claude Sonnet 4-6	**1.876**	**0.759**	**0.068**	×
LLaMA-3.3-70B	5.291	0.259	0.103	×

Table 5: Demographic Sensitivity: Independent-Samples t-tests (RQ3). ✓ = significant at α=0.05; × = not significant.

Model	Age p	Gender p	Exp. p
Human	0.831	0.722	<0.001 ✓ (t=6.92)
GPT-3.5	0.242	0.119	0.341
LLaMA-2	0.330	1.000	0.665
GPT-4o	0.964	0.519	0.870
Claude 4-6	0.416	0.403	0.979
LLaMA-3.3	1.000	0.551	0.689

most broadly approved response to any given question; for professional survey questions about developer practices, this is consistently the “best practice” answer (e.g., use version control, prefer typed languages, communicate proactively), causing all RLHF-aligned models to converge on the same handful of choices regardless of the demographic profile presented.

4.2 RQ1 — Chi-Square Distributional Similarity

Table 4 presents chi-square tests comparing each model’s response distribution to the GPT-3.5 P4 few-shot reference distribution.

The *Distribution Paradox* emerges here. Phase 2 models are simultaneously *more uniform* (higher κ from Table 3) and *more similar to the reference human distribution* (lower χ^2, higher p) than Phase 1 models. This apparent contradiction resolves when one recognizes that RLHF trains models toward the single most-approved response for each question, which also happens to be the most frequently observed human response. The result is models that are internally consistent (same answer across all demographic profiles) and externally well-calibrated (that same answer is the modal human answer)—but that *cannot* reproduce the distribution of *minority* responses that make human data valuable. Claude Sonnet 4-6 achieves the smallest divergence (V=0.068), possibly reflecting Anthropic’s Constitutional AI approach producing stronger regression to modal human responses than pure RLHF.

4.3 RQ3 — Demographic Sensitivity (t-tests)

Table 5 presents t-test results across all three demographic dimensions for each respondent type.

Profile blindness is complete and consistent across both generations. No model in either phase shows statistically significant response differences on any demographic dimension. The only significant demographic effect in the dataset belongs to human respondents: experienced developers (≥3 yr) systematically choose different answers to career-stage-sensitive questions than those with <1 yr experience (t=6.92, p<0.001).

Phase 2 models are, if anything, *more* profile-blind than Phase 1: GPT-4o’s Experience p-value (0.870) exceeds GPT-3.5’s (0.341), and Claude’s Experience p=0.979 is the highest in the entire table. Constitutional AI and extensive RLHF tuning appear to enforce stronger persona-uniformity than the lighter alignment of Phase 1 models. Crucially, this finding holds at an N where medium-to-large demographic effects ($d \geq 0.5$) would be reliably detected (see §3.7).

Table 6: One-Way ANOVA: Cross-Model Response Differences (RQ3). Note: these F-statistics measure differences *between* respondent types (human vs. each LLM), not within-model demographic sensitivity. They are not contradicted by the t-test null results.

Question	F	p	
Biggest challenge (Q3)	41.825	<0.001	✓
Language selection (Q4)	102.956	<0.001	✓
Innovation vs. deadline (Q7)	236.796	<0.001	✓

Table 7: Qasper Automatic Metrics (50 papers, 124 QA pairs, avg. P1–P4 few-shot). Bold = best within Phase 2. TF-IDF Sim‡ is not comparable to Phase 1 DeBERTa BERTScore.

Model	B-1	B-4	R-1	R-L	MTR	TF-IDF‡
GPT-3.5 (Ph. 1)	0.421	0.187	0.389	0.367	0.298	—
LLaMA-2 (Ph. 1)	0.298	0.107	0.301	0.282	0.223	—
GPT-4o	0.364	0.206	0.307	0.298	0.449	0.255
Claude 4-6	**0.405**	**0.232**	**0.312**	0.301	**0.523**	**0.318**
LLaMA-3.3	0.445*	0.249*	0.413*	**0.395***	0.507	0.271

*LLaMA-3.3’s extractive style matches gold span vocabulary, inflating BLEU/ROUGE.
‡TF-IDF cosine similarity; not comparable to Phase 1 DeBERTa BERTScore.

4.4 RQ3 — ANOVA (Cross-Model Differences)

All three ANOVA tests are highly significant (Table 6), confirming that *respondent type* (human vs. each LLM) produces measurably different response patterns. The large F-statistics reflect the substantial qualitative differences in how human developers and LLMs frame professional challenges, language selection, and deadline trade-offs. This is *orthogonal* to the t-test results: ANOVA asks whether GPT-4o responds differently *from humans*, while t-tests ask whether GPT-4o-as-22-year-old-woman responds differently from GPT-4o-as-35-year-old-man. A model can fail both (as all LLMs do here).

4.5 RQ2 — Qasper Automatic Metrics

Unlike the survey simulation task, the Qasper benchmark shows consistent, substantive improvements for Phase 2 models (Table 7). METEOR—the metric most robust to paraphrase and synonym variation—improves by +0.151 for GPT-4o (0.449 vs. 0.298) and by +0.225 for Claude (0.523 vs. 0.298), a 75% relative gain. LLaMA-3.3-70B leads on BLEU and ROUGE, reflecting its concise, extractive answer style

Table 8: Qasper Stylistic Metrics (avg. P1–P4 few-shot).§

Model	Relev.	Fluency	MTLD	Flesch	Correct.
GPT-3.5§	0.612	—	68.4	52.1	0.641
LLaMA-2§	0.487	—	45.2	48.3	0.502
GPT-4o	0.128	0.896	24.1	31.4	0.658
Claude 4-6	0.125	0.851	**35.0**	31.6	**0.762**
LLaMA-3.3	**0.135**	0.861	33.0	**33.0**	0.741

§**Critical note:** Phase 1 relevance uses sentence-transformer cosine; fluency is GPT-2 perplexity (lower = better). Phase 2 uses TF-IDF cosine and TTR (higher = better). These columns are **not** directly comparable across phases. Only within-Phase 2 comparisons are valid.

Table 9: Per-Prompt Qasper Performance (Phase 2, BLEU-4 / METEOR). P5 survey simulation results are pending completion of inference runs.

Prompt	GPT-4o	Claude 4-6	LLaMA-3.3
P1 (Expert NLP)	0.243 / 0.459	0.296 / 0.556	0.306 / 0.526
P2 (Plain QA)	0.153 / 0.368	0.201 / 0.512	0.254 / 0.537
P3 (Concise)	0.152 / 0.468	0.161 / 0.499	0.143 / 0.461
P4 (Novice)	0.277 / 0.500	0.271 / 0.524	0.294 / 0.501
P5 (CoT) survey simulation: inference pending			

that directly matches gold span vocabulary. GPT-4o's ROUGE-L appears to decline relative to GPT-3.5 (0.298 vs. 0.367); this is an artifact of GPT-4o's more elaborate, multi-sentence answers being penalized by ROUGE-L's longest common subsequence measure—its METEOR improvement (+51%) confirms genuine semantic quality gains.

4.6 RQ2 — Qasper Stylistic Metrics

Within the Phase 2 cohort, Claude Sonnet 4-6 leads on correctness (0.762) and formality (MTLD = 35.0), consistent with its strong METEOR performance. LLaMA-3.3-70B achieves the highest relevance score (0.135) and best readability (Flesch = 33.0). GPT-4o shows the highest lexical variety (TTR = 0.896), reflecting its more verbose, paraphrase-rich answer style.

4.7 Prompt Variant Analysis

P1 (Expert NLP) and P4 (Novice) consistently outperform P2 (Plain QA) and P3 (Concise) across all Phase 2 models on both BLEU-4 and METEOR (Table 9). P2 produces the weakest BLEU-4 performance for GPT-4o (0.153) and Claude (0.201), suggesting that minimal framing leads to answers that match gold spans less well despite reasonable METEOR scores. P3 performs inconsistently: its conciseness instruction produces strong METEOR for GPT-4o (0.468) but poor BLEU-4 (0.152), reflecting lexically diverse but span-divergent responses. LLaMA-3.3-70B maintains competitive BLEU-4 even under P2 (0.254), consistent with its extractive answer style being relatively insensitive to prompt framing.

5 DISCUSSION

5.1 The Kappa Paradox: RLHF Homogenizes Synthetic Respondents

The *Kappa Paradox* is the central theoretical contribution of this study. Naïvely, one would expect that more capable, better-aligned 2026 models would produce responses *closer* to the human baseline than their 2023 predecessors. Instead, Phase 2 models are further away by every Fleiss' Kappa measure: $\Delta\kappa_{\text{human}}$ = +0.377 to +0.384 for Phase 2 vs. +0.135 to +0.139 for Phase 1.

The mechanism is RLHF's optimization objective. Human raters reward responses that are professionally appropriate, widely applicable, and socially acceptable. For a 12-question developer survey, the "best" answer to "What is your biggest challenge?" is almost always something like "Keeping up with rapidly changing technology"—not because it is the most demographically authentic answer, but because it is the most likely to satisfy a general human evaluator. An RLHF-trained model, regardless of the demographic profile it is asked to inhabit, converges on this response because doing otherwise would have been penalized during training.

The convergence of three models from *three different organizations* to $\kappa \in [0.480, 0.487]$ (span = 0.006) is the strongest evidence for this mechanism. GPT-4o, Claude Sonnet 4-6, and LLaMA-3.3-70B share neither architecture, training data, nor fine-tuning methodology—except the RLHF paradigm itself. Model architecture cannot explain a convergence this tight; only the shared training objective can.

The practical implication is counterintuitive but important: *a more helpful, better-aligned LLM is a worse synthetic survey respondent.* Researchers who select the most capable available model for survey simulation tasks may inadvertently select the one most pathologically unable to reproduce demographic diversity.

5.2 The Distribution Paradox: Uniform Yet Calibrated

The *Distribution Paradox* appears to contradict the *Kappa Paradox*. If Phase 2 models give the same answer regardless of demographic profile (high κ), how can they simultaneously *match* the human response distribution (low χ^2)?

The resolution is that the single answer Phase 2 models converge on is the *modal human answer*—the response chosen by a plurality of the 314 human respondents. RLHF training essentially distills the "average approved human response" into the model's output distribution. For a survey population of working developers, this modal response happens to be the professionally expected one, and it also happens to be the most common one. The result is a model that looks well-calibrated when evaluated at the distributional level (low χ^2) but is profoundly miscalibrated at the individual level (high κ): it can tell you what most developers would say, but cannot tell you what a 27-year-old Asian woman with < 1 year of experience would say differently from a 22-year-old white man with 5 years of experience.

Claude's smallest divergence (V=0.068) and highest kappa (κ=0.4865) jointly exemplify this paradox most sharply: Constitutional AI may be particularly effective at learning the modal human response while simultaneously reducing the variance needed for authentic demographic simulation.

5.3 Profile Blindness: A Persistent Cross-Generation Failure

The *profile blindness* finding is striking in its consistency. Both Phase 1 and Phase 2 models fail to produce statistically significant response differences across Age, Gender, and Experience—covering a broad range of model sizes (7B to ~200B parameters), training objectives (SFT, RLHF, Constitutional AI), and organizations.

The failure is not merely "no improvement from Phase 1 to Phase 2": Phase 2 models are measurably *worse* at demographic differentiation. Claude's Experience p=0.979 versus GPT-3.5's p=0.341, and LLaMA-3.3's Age p=1.000 versus LLaMA-2's p=0.330, indicate that scale and alignment have increased, not reduced, persona homogenization.

This has a critical implication for practitioners: **the absence of significant demographic t-tests is not evidence of unbiased, accurate simulation; it is evidence of profile blindness.** A model that ignores demographic prompts entirely would also produce non-significant demographic t-tests. The two outcomes (successful parity, profile blindness) are statistically indistinguishable by t-test alone and must be triangulated with κ and distributional analysis.

5.4 Qasper: Genuine Capability Improvements

Unlike the survey simulation task—where RLHF appears systematically detrimental—the Qasper scientific QA benchmark benefits from Phase 2 scale and alignment. METEOR improvements are dramatic and consistent: +51% for GPT-4o (0.449 vs. 0.298), +75% for Claude (0.523 vs. 0.298), and +127% for the LLaMA lineage (0.507 vs. 0.223). These are improvements in a task where "giving the most broadly approved response" aligns with factual accuracy—unlike demographic survey simulation, where it actively undermines authentic persona rendering.

Two caveats apply. First, the TF-IDF proxy values (0.255–0.318) cannot be compared to Phase 1 DeBERTa BERTScore values; any comparison would be misleading. Second, the stylistic metrics cross-phase comparison is similarly invalid (different underlying methods), and only within-Phase 2 rankings should be interpreted.

5.5 Practical Guidance for SE Researchers

Based on the joint evidence from κ, χ^2, and t-tests, we recommend the following checklist before deploying any LLM as a synthetic survey respondent pool:

(1) **Verify Fleiss' Kappa proximity.** Compute within-group Fleiss' κ for the LLM responses and confirm it falls within ±0.05 of the expected human diversity level for your target population. $\kappa > 0.35$ with a diverse human baseline near $\kappa \approx 0.10$ is a red flag.

(2) **Run chi-square distributional similarity tests.** Non-significant χ^2 (p > 0.05) against the human distribution is necessary but not sufficient—the *Distribution Paradox* shows models can pass this test while still being profile-blind.

(3) **Interpret absent demographic t-tests as profile blindness.** If no demographic dimension (Age, Gender, Experience) yields a significant t-test, treat this as evidence that the model is ignoring demographic prompts—not as evidence of accurate, unbiased simulation.

(4) **Use LLMs for instrument validation, not demographic substitution.** LLMs are appropriate for piloting questionnaire wording, identifying ambiguous items, and stress-testing response categories. They are not appropriate substitutes for demographically diverse human participants when the research question involves demographic variation.

6 LIMITATIONS

L1 — Profile coverage. The ten synthetic profiles exclude developers over 35, non-binary gender identities, and respondents from Global South geographic and cultural contexts, limiting generalizability.

L2 — Single domain. Results are specific to a software developer survey about professional practices. Generalization to other qualitative research domains (health, social science, education) requires separate validation.

L3 — Statistical power. With N=120 per condition (10 profiles × 12 questions), the study reliably detects medium-to-large demographic effects ($d \geq 0.5$) but is underpowered for small effects ($d < 0.3$, $\beta < 0.55$). All null demographic sensitivity results should be interpreted as "no medium-to-large effect detected."

L4 — API non-determinism. Despite temperature = 0.7 and seed = 42, cloud API responses may differ across API versions or infrastructure updates, limiting exact reproducibility over time.

L5 — Groq rate limits. LLaMA-3.3-70B inference required rotation across eight Groq API accounts due to daily token-per-day limits. Inference was conducted in serial batches; minor cross-batch inconsistencies cannot be ruled out.

L6 — Metric proxies. System RAM constraints (8 GB fully utilized) prevented loading DeBERTa (~1.5 GB) for BERTScore and GPT-2 for perplexity. TF-IDF cosine similarity and TTR are scientifically reasonable proxies but should not be treated as equivalent to their ML-based counterparts.

L7 — P5 survey simulation. Chain-of-Thought (P5) survey simulation inference is ongoing at time of submission. Results will be incorporated in the camera-ready version; a preliminary kappa value will be reported in the author response if available.

L8 — No LLM-as-judge evaluation. A planned GPT-4o-as-judge evaluation on 100 randomly sampled Qasper pairs was omitted due to cost constraints. Stylistic metric ratings rely solely on reference-based automatic scorers.

7 CONCLUSION

We presented the first complete longitudinal evaluation of LLMs as synthetic survey respondents, comparing Phase 1 models (GPT-3.5-Turbo, LLaMA-2-7B, 2023) with Phase 2 successors (GPT-4o, Claude Sonnet 4-6, LLaMA-3.3-70B, 2026) on an identical protocol against a n=314 human developer baseline.

RQ1 (survey simulation fidelity). RLHF-aligned 2026 models converge to $\kappa \in [0.480, 0.487]$—nearly twice the Phase 1 values and four times the human baseline. The *Kappa Paradox* is confirmed:

higher alignment worsens demographic simulation. The complementary *Distribution Paradox* shows these same models match human response distributions more closely ($\chi^2 < 5.3$, $p > 0.25$), because RLHF learns the modal human response, not the distributional diversity of human responses.

RQ2 (Qasper quality). Genuine capability improvements are observed across all three Phase 2 models, with METEOR gains of +51% to +75% over Phase 1 baselines and correctness gains of +2.7% to +18.9%. LLaMA-3.3-70B leads on BLEU and ROUGE; Claude Sonnet 4-6 leads on METEOR and correctness.

RQ3 (demographic sensitivity). Complete *profile blindness* holds across both generations. No model from either phase shows any significant demographic sensitivity across Age, Gender, or Experience. Phase 2 models show even higher Experience p-values than Phase 1, suggesting that Constitutional AI and extended RLHF tuning strengthen, rather than ameliorate, profile blindness.

Future work should extend this longitudinal framework to multimodal models, broader geographic and demographic profiles, retrieval-augmented generation for scientific QA, and—most importantly—evaluation protocols designed specifically to measure *demographic diversity* rather than response consistency, providing the missing instrument for assessing whether any LLM can genuinely function as a demographically differentiated synthetic respondent.

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